

Assignment - 03

Section - A

1. What is an association rule?

Ans. An association rule shows a relationship b/w two itemsets in the form $X \rightarrow Y$. It indicates that the presence of item X in a transaction implies the presence of item Y .

2. Define frequent pattern.

Ans. A frequent pattern is an itemset, sequence or structure that appears frequently in a data set. It satisfies the minimum support threshold.

3. What does iterative approach mean in Apriori algo?

Ans. Apriori uses an iterative method of generating $(k+1)$ -itemsets from k -itemsets. Repeat until no more frequent itemsets can be found.

4. Write applications of association rules.

Ans. • Market basket analysis for product purchase pattern.
• Used in cross-selling, web usage mining, fraud detection and recommendation systems.

5. Label the association rule: $X \rightarrow Y$ [Support 50%, Confidence 70%]

- Ans. • Support 50% $\rightarrow$ X & Y occur together in 50% of total transaction.
- Confidence 70% $\rightarrow$ When X occurs, 70% chances that Y also occurs.

Section-B.

1. Define support & confidence in relation to frequent patterns.

Ans. • Support:

- $\rightarrow$ Measures how frequently an itemset occurs in the dataset.
- $\rightarrow \text{Support}(X \rightarrow Y) = \frac{\text{Transactions containing } X \ \& \ Y}{\text{Total Transactions}}$
- $\rightarrow$ Indicates the overall importance of a rule.

• Confidence:

- $\rightarrow$ Measures accuracy of the rule.
- $\rightarrow \text{Confidence}(X \rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X)}$
- $\rightarrow$ Shows how often Y appears when X appears.

• Eg:

→ If 20 out of 100 transactions contain bread & butter
Support = 20%.

→ If 20 out of 25 bread transactions contain butter:
Confidence = 80%.

- Importance: Helps evaluate the strength and usefulness of association rules.

2 Briefly describe correlation analysis.

- Ans:
- Determines how strongly items are related in itemsets.
 - Helps check whether association rules show real relationships or are due to chance.
 - Uses lift, chi-square & conviction to measure correlation.
 - Positive correlation: Occurrence of one item raises the likelihood of another.
 - Negative correlation: Items rarely occur together despite high individual support.
 - $\text{Lift} > 1 \rightarrow$ Positive correlation
 $\text{Lift} < 1 \rightarrow$ Negative
 - Ensures more reliable rules beyond support-confidence.

3. Explain market basket analysis and its uses.

Ans. • Technique to analyse customer purchase behaviour.
• Finds relationships b/w purchased items using association rules.

• Eg: Customers buying bread often buy butter.
• Uses:

- Product placement strategies in stores.
- Cross-selling & upselling.
- Designing combo offers & promotions.
- Inventory & shelf management.
- Personalized recommendation systems in e-commerce.

Section-C

1. Explain association rules with the help of Apriori algo using suitable eg.

Ans. • Association rules: Discover relationships of type $X \rightarrow Y$.
• Apriori algo: Used to find frequent itemsets and generate rules.

• Steps of Apriori:

- i) Set minimum support and confidence.
- ii) Generate 1-itemsets, find frequent ones.
- iii) Generate 2-itemsets from frequent 1-itemsets, check

Support.

- iv) Repeat until no more frequent itemsets.
- v) Generate rules from frequent itemsets using confidence.

• Example Transaction Datasets:

T1: {Milk, Bread}

T2: {Milk, Diaper, Beer}

T3: {Bread, Diaper, Butter}

T4: {Milk, Bread, Diaper}

T5: {Bread, Butter}

• Frequent itemsets found using support threshold.

• Eg rule: {Milk, Bread} $\rightarrow$ Diaper.

• Support = % of transactions containing all 3 items.

• Confidence = $\frac{\text{Support (Milk + Bread + Diaper)}}{\text{Support (Milk + Bread)}}$

• Conclusion: Helps in product placement and customer behaviour understanding.

2. Briefly explain FP-Growth Algorithm using suitable example.

Ans: • FP-Growth stands for Frequent Pattern Growth.
• Used to find frequent patterns without candidate generation.

- Steps:

- i) Scan dataset to find item frequencies.
- ii) Build FP-tree (compressed form of transactions)
- iii) Generate conditional FP-trees for items.
- iv) Extract frequent patterns.

- Example:

- Build FP-tree from dataset based on descending frequency.
- Mine tree to get patterns like {Bread, Milk}, {Diaper, Beer}.

- Advantages:

- Faster than Apriori
- less memory usage
- Suitable for large datasets.

3. Define Normalization. Also discuss

(a) Normalization Min-Max.

(b) Z-score Normalization.

Ans. • Normalization

- Process of scaling data into a smaller range to improve accuracy of ML models.

→ Removes bias due to different measurement units.

a) Min-Max Normalization:

→ Scales values into a fixed range (usually 0 to 1)

→ Formula: $(v - \min) / (\max - \min)$

→ Preserves relationships b/w original values.

b) Z-score Normalization:

→ Based on mean and standard deviation.

→ Formula: $(v - \text{mean}) / \text{SD}$.

→ Converts values into no. of standard deviations from mean.

→ Useful for data out outliers.